International Journal of Innovation Management
Vol. 22, No. 7 (October 2018) 1850059 (22 pages)

DOI: 10.1142/S1363919618500597

DOES STRONGER R&D CAPABILITY ALWAYS PROMOTE BETTER INNOVATION? THE MODERATING ROLE OF KNOWLEDGE BOUNDARY SPANNING OF R&D NETWORK

YANHUI JIANG*, CHONGYANG WEI*, ZHI YANG* and
ULAGANATHAN SUBRAMANIAN*,†,‡

Business School, Hunan University, Changsha, China
†*UTB Business of School, Universiti Teknologi Brunei, Brunei*
‡*nath.subra@utb.edu.bn*

Published 8 February 2018

Setting the research background in China, this study draws on absorptive capacity, knowledge inertia and prospect theory to show the relationship between R&D capability and innovation performance which comprises exploitation and exploration. We propose that stronger R&D capability promotes exploitation performance but inhibits exploration performance. As exploitation represents immediate interest and exploration represents long-term interest, we introduce the notion of knowledge boundary spanning of R&D network to balance short-term and long-term benefits. The empirical results show that R&D capability and knowledge boundary spanning of R&D network complement each other for explorative innovation while they present trade-offs for exploitative innovation. This study contributes to existing literature on R&D capability–innovation performance, and it further extends our understanding by investigating the impact of knowledge boundary spanning of R&D network on the R&D capacity–innovation performance relationships. In addition, this study provides references on resources configuration to achieve different innovation strategies.

Keywords: Research and development; innovation; China; knowledge boundary.

Introduction

Innovation is a key strategy for organisations' survival and development especially in developing countries (Ooi *et al.*, 2012). The Chinese government has been encouraging people to start their own businesses and to make innovations. Chinese

‡Corresponding author.

firms have been investing a lot on R&D capability. However, does stronger R&D capability always promote better innovation performance?

According to Paruchuri and Eisenman (2012) R&D capability is a kind of capability that recombines existing knowledge in a novel way to generate new knowledge and innovative outputs (Paruchuri and Eisenman, 2012). With R&D capability, firms extend the existing R&D outputs to generate new R&D outputs through exploitation, exploration and application of prior knowledge and new knowledge. R&D capability covers the direction and intensity of technology development (Yong, 2011). There are numerous literatures on R&D capability. Yong (2011) adopts Cobb–Douglas production function to demonstrate that R&D capability has a positive impact on innovation performance of high technology enterprises. As the enterprises' sizes increase, R&D capability impacts innovation performance at an accelerating speed. Paruchuri and Eisenman (2012) study the R&D network by investigating the micro foundations of R&D capability, analysing innovation strategy choices and innovation outputs from perspectives of individual capability and organisation choices.

According to the Oslo Manual prepared by the Organisation for Economic Cooperation and Development (OECD, 1997), innovation can be divided into product innovation, improved product innovation and process innovation. Boer and During (2001) added another type of innovation, organisational innovation. Gunday *et al.* (2011) proposed a fourth type of innovation, marketing innovation. Ayhan (2011) stated that new uses of existing knowledge and technology as well as creating knowledge through research would lead to innovation. In this paper, we will focus on product innovation. According to March (1991) and Zhou and Wu (2010), innovation can be divided into exploitative innovation and explorative innovation. Exploitative innovation which is incremental, mainly exploits, renews and improves existing knowledge and technologies, aiming at organisations' short-term development. Explorative innovation, on the other hand, is disruptive and radical, mainly explores and digs into completely new knowledge and technologies, aiming at organisations' long-term development.

With various theories and different economic environments, existing literatures indicate different kinds of relationships between R&D capability and innovation performance. Building on absorptive capacity theory, Rosenkopf and Nerkar (2001) argued that the diversity of capabilities promote technology exploration. As an organisation builds stronger R&D capability, its receptivity to external information is higher and its exploration is better. Rooted in knowledge-based view, Xu *et al.* (2013) indicate that the internal technological strength has inverted U-shaped relationships with both exploitative and explorative innovations. A higher level of knowledge base results in greater inertia, thus weakening the receptivity and transformation of external new knowledge, and ultimately hinders exploration performance. As organisations accumulate more knowledge, there will be a higher level of knowledge diversity and a lower level of knowledge relevance which inhibits the progress of exploitation performance. Zhou and Wu (2010) studied the influence of strategy choice and individual inclination on exploitation and exploration. Bounded by the sunk cost of existing knowledge and uncertainty of explorative innovation, organisations with stronger R&D capability tend to carry out exploitative innovation rather than explorative innovation.

Setting the economic background in China, this study investigates the relationship between R&D capability and innovation performance which includes exploitative innovation and explorative innovation. We propose that R&D capability promotes both kinds of innovation from the view of absorptive capacity theory (Cohen and Levinthal, 1990). However, the results are the opposite for exploitative performance and explorative performance from the view of knowledge inertia (Liao, 2002) and prospect theory (Kahneman, 1979).

According to Liao *et al.* (2008), there are two dimensions of knowledge inertia, experience inertia and learning inertia. The presentation of experience inertia (Liao *et al.*, 2008) is solving problems by utilising prior experience and knowledge, thus reinforcing organisations to depend on current R&D trajectories and it tends to generate more incremental innovation outputs (Liao, 2002; Yang *et al.*, 2014). Learning inertia means that the current knowledge systems inhibit learning of new knowledge, while explorative innovation requires organisations to assimilate and transform new knowledge (Yang *et al.*, 2014). Therefore, we propose that stronger R&D capability promotes exploitation but inhibits exploration.

According to prospect theory (Kahneman, 1979), organisations are risk-averse in the face of certain returns. In addition, organisations with excellent innovation performance tend to leap from exploration to exploitation. So, we propose that organisations with stronger capability carry out exploitation rather than exploration.

To sum up, we hypothesise that R&D capability has a positive impact on exploitation performance and an inverted U-shaped relationship with exploration performance.

As exploitation and exploration are different kinds of innovation unlike commercialisation attributes, they depict different strategy developments. Exploitation corresponds to immediate returns and exploration is essential to long-term strategy. In other words, the equilibrium of exploitation and exploration are meaningful for both short-term and long-term developments (Levinthal and March, 1993). To achieve the balance of exploitation and exploration, we introduce knowledge boundary spanning of the R&D network (Yang *et al.*, 2013) as moderator, and investigate its moderating effects on the relationships between R&D capability and innovation performance.

According to Tijssen (1998), and Yang (2014), R&D network is an evaluative and interdependent system based on resources' relationships, the characteristics of this system is interaction, process, procedure and institutionalisation. Boundary spanning is an important topic of studies on network (Rosenkopf and Nerkar, 2001). In Yang (2014) and Yang *et al.* (2013), they investigated the influences of knowledge boundary spanning of R&D network on innovation catching-up. Knowledge boundary spanning includes the spanning of knowledge breadth and knowledge depth. The former means more kinds of knowledge from more knowledge domains while the latter means the augmenting of knowledge complexity and specialisation. Additionally, the meaning of breadth spanning conforms to the attributes of exploration while depth spanning conforms to the attributes of exploitation. Breadth spanning inhibits organisations' learning inertia while depth spanning promotes organisations' experience inertia. Therefore, from the view of knowledge inertia theory, knowledge boundary spanning of R&D network negatively moderates the relationship between R&D capability and exploitative performance, but positively moderates the relationship between R&D capability and explorative performance.

According to prospect theory (Kahneman, 1979), organisations tend to choose ways with less risk facing certain returns. With the augmenting of knowledge depth spanning, R&D teams' capability to comprehend and utilise knowledge of existing domains gets enhanced, thus they tend to perform incremental and exploitative innovation for less risk-taking and with more certain returns. R&D capability promotes exploitative innovation outputs through enhancement of experience inertia and risk-averse tendency, while knowledge boundary spanning of R&D network promotes exploitative innovation with the same mechanisms. Therefore, we believe that it is a waste of resources to invest heavily in both R&D capability and knowledge boundary spanning of R&D network.

On the contrary, with the augmenting of knowledge breadth spanning, R&D teams gain stronger capability to adjust aspirations of different kinds of innovations timely, especially in paying more attention to the long-term returns from explorative innovation. R&D teams' risk-averse tendency will be weakened and there will be better explorative outputs. As for R&D teams with stronger R&D capability, explorative innovation outputs will be inhibited through learning inertia and risk-averse tendency. However, knowledge boundary spanning of R&D network inhibits R&D teams' learning inertia and risk-averse tendency. Furthermore, knowledge boundary spanning of R&D network positively moderates R&D capability–exploration performance relationship. For explorative innovation, we believe that there is complementary relationship between R&D capability and knowledge boundary spanning of R&D network.

In sum, we propose that knowledge boundary spanning of R&D network negatively moderates the relationship between R&D capability and exploitative performance, while positively moderates the relationship between R&D capability and explorative performance. Contributing to existing literatures, this study examines the role of R&D capability in exploitative innovation and explorative innovation, and creatively introduces the notion of knowledge boundary spanning of R&D network as moderator. More importantly, we offer a contingency perspective by demonstrating different influences of various resources on exploitative performance and explorative performance. As a result, this study provides organisations with suggestions on resources configuration to achieve equilibrium innovation or some kind of innovation strategy.

Theory and Hypotheses

Relationship between R&D capability and exploitative innovation performance

Organisations adept in certain technology domains tend to acquire knowledge similar to existing knowledge base in order to gain innovation outputs and innovation benefits efficiently (Levinthal and March, 1993). As Cohen and Levinthal (1990) argued, R&D investments can not only generate new knowledge, but also improve organisations' absorptive capacity. Under these circumstances, R&D staff with accumulation of technologies have a better understanding of existing knowledge domains and current technological trajectories. Thus, R&D capability reinforces exploitation of current knowledge and technologies from the perspective of absorptive capacity theory (Cohen and Levinthal, 1990).

Building on knowledge-based view as the knowledge accumulation of current technological domains, organisations' capabilities to grasp and transform knowledge or experience of local domains are stronger. With the influence of self-reinforcement attribute, organisations tend to integrate similar knowledge into its knowledge base (Levinthal and March, 1993; Zhou and Wu, 2010). Besides, the existence of knowledge inertia (Liao, 2002) intensifies organisations' knowledge and experience of current technological domain. This is specially the case for experience inertia (Liao *et al.*, 2008) included in knowledge inertia which means solving problems utilising prior experience and knowledge. The enhancement of current knowledge and experience promotes exploitative innovation. In other words, through knowledge inertia theory, stronger R&D capability reinforces organisations to depend on current R&D trajectories and tends to generate more incremental innovation outputs (Liao, 2002; Yang *et al.*, 2014). Therefore, stronger R&D capability promotes better exploitative innovation performance.

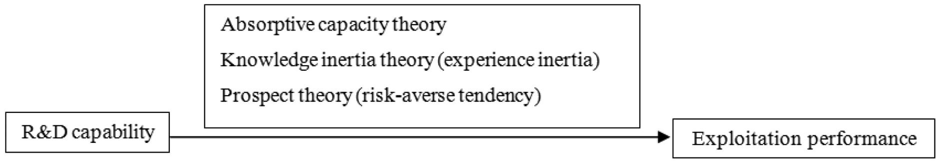

Fig. 1. Theoretical framework of R&D capability — exploitation performance.

According to the prospect theory (Kahneman, 1979), organisations are risk-averse in the face of certain returns. More concretely, organisations with stronger R&D capability have gained excellent R&D outputs or returns, and moderate success will decrease organisations' risk preference level. Furthermore, exploitation which is incremental, generates efficient returns with less uncertainty. Thus, organisations with stronger R&D capability tend to generate better exploitative performance (Levinthal and March, 1993; Zhou and Wu, 2010; March, 1991).

Figure 1 shows the theoretical framework of R&D capability–exploitative performance relationship, and we propose that:

Hypothesis 1a: *R&D capability has a positive impact on exploitation performance.*

Relationship between R&D capability and explorative innovation performance

Organisations often invest substantially in R&D which include accumulation of new knowledge, discoveries of new opportunities and training for R&D staff to build internal R&D strength (Afuah, 2002). As a result, organisations develop stronger R&D capability to recognise technology tendencies and cross-technology boundaries (Rosenkopf and Nerkar, 2001). According to absorptive capacity theory, if an organisation's ability in problem solving is stronger, its capability to assimilate and generate new knowledge is stronger (Cohen and Levinthal, 1990). The above findings conform to attributes of exploration so that accumulations of R&D capability could promote explorative outputs.

From the perspective of knowledge inertia theory and prospect, there is inhibition rather than promotion of stronger R&D capability to explorative innovation performance. Along with the augmenting of R&D capability, accumulated knowledge base of corresponding domains is more entrenched, and it is more difficult to assimilate completely new knowledge into the current knowledge base (Dougherty and Heller, 1994; Zhou and Wu, 2010). Furthermore, the existence of knowledge inertia reinforces organisations to rely on current R&D trajectories and resist the assimilation and transformation of new knowledge and technologies to some degree. Learning inertia included in knowledge inertia means that current

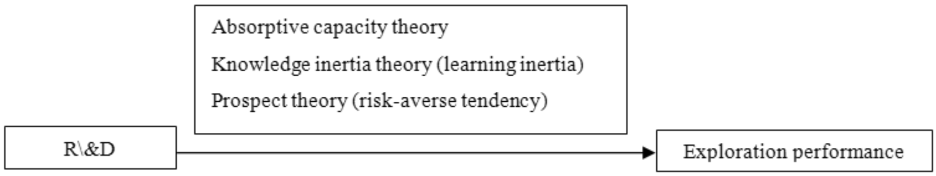

Fig. 2. Theoretical framework of R&D capability — exploration performance.

knowledge system inhibits learning of new knowledge. Unfortunately, explorative innovation requires organisations to break current technological trajectories with novel thinking patterns, assimilate new knowledge, transform new knowledge and innovate disruptively (Yang *et al.*, 2014). Thus, there is inhibition of stronger R&D capability to explorative innovation.

In accordance with prospect theory, (Swift, 2016) point out that when innovation performance is better than aspirations, organisations tend to leap from exploration to exploitation even though there are promising opportunities if organisations were to continue exploration. Ascompared with exploitation, exploration has more difficulties in the integration of knowledge base, higher standard for innovation, larger uncertainty and longer commercialisation process, thus exploration means higher risk for investors (Levinthal and March, 1993; Rui, 2016). It is clear that organisations with stronger R&D capability have achieved excellent innovation performance. They tend to carry out exploitation rather than exploration facing certain returns. There is inhibition of stronger R&D capability to explorative innovation.

To sum up, from the perspective of absorptive capacity theory, knowledge inertia theory and prospect theory, we posit that the relationship between R&D capability and explorative innovation performance is inverted, U-shaped. That is to say when explorative innovation performance achieves the optimal point, it will decrease while R&D capabilityincreases.

Figure 2 shows the theoretical framework of R&D capability–exploration performance, and we propose that:

Hypothesis 1b: *R&D capability has an inverted U-shaped relationship with exploration performance.*

Knowledge boundary spanning of R&D network

From the view of knowledge inertia and prospect theory, stronger R&D capability promotes exploitative outputs but inhibits explorative innovation outputs. Exploitation conforms to corporations' short-term developments and long-term benefits while exploration aims at corporations' long-term development and long-term returns. However, equilibrium of exploitation and exploration is better for firms' short-term survival and long-term development (Levinthal and March, 1993). We introduce the notion of knowledge boundary spanning of R&D network (Yang *et al.*, 2013) to examine its moderating role in R&D capability–exploitation performance relationship and R&D capability–exploration performance relationship to help firms achieve both short-term survival and long-term development.

According to the viewpoints of Tijssen (1998) and Yang (2014), R&D network is an evaluative and interdependent system based on resources' relationships and the characteristics of the system are interaction, process, procedure and institutionalisation. From the viewpoint of Meng (2015), R&D staff from one company constitutes the R&D staff knowledge network.

As for dimensions, knowledge boundary spanning includes knowledge depth spanning and knowledge breadth spanning. Existing literatures mainly concentrate on boundary spanning from Inside Company to outside but intra-organisational boundary spanning gets less attention (Rosenkopf and Nerkar, 2001). There are shortcomings to cross-inter-organisational boundaries. According to absorptive capacity theory, shared knowledge and experience are essential for communication and effective communication cannot be guaranteed without enough knowledge overlap (Cohen and Levinthal, 1990). Still and Strang (2009) argued that connects with social proximity are in favour of knowledge utilisation which influences processes of knowledge generation and product innovation. Tortoriello and Krackhardt (2010) show that if staff from different organisations with strong ties share common knowledge, boundary spanning is strongly associated with the generation of innovation. On the contrary, it is difficult to absorb knowledge across boundaries without shared knowledge to support. Besides, network with strong ties enable diffusion of tacit knowledge to increase innovation efficiency (Seibert *et al.*, 2001). Intra-organisation is a natural environment providing relationships with strong ties, yet we know little about intra-organisational knowledge boundary spanning of R&D network. This study investigates the moderating role of intra-organisational knowledge boundary spanning of R&D network. However, there is a more explicit question: What "knowledge boundary spanning of R&D network" is actually inside the company? Due to the high complexity and specialisation of R&D activities, intra-organisational knowledge boundary spanning of R&D network aims at assimilation, transformation and application of knowledge from skilled staff. According to existing literatures, different from R&D human capital, skilled human capital is another kind of innovation human capital (Xianxiang, 2009). Skilled staff possess plenty of expertise and experience mainly working on operation, maintenance and so on, and transform R&D outputs to practical productivity. For example, software test staff in software companies are skilled staff. Furthermore, skilled human capital is also an important source of R&D innovation (Xianxiang, 2009). Zheng (2013) demonstrated that skilled human capital has a positive impact on exploitative innovation but has no significant impact on explorative innovation while skilled human capital of high level promotes explorative innovation. Taking previous literatures as references, we take skilled staff as proxy for knowledge boundary spanning of R&D network and examine its moderating role in R&D capability–innovation performance.

The moderating role of knowledge boundary spanning of R&D network on R&D capability–exploitation performance relationship from the vertical dimension.

Knowledge boundary spanning of R&D network includes the spanning of knowledge depth which means uniqueness and complexity of knowledge content (Zhou and Li, 2012). As knowledge attribute of path-dependence, the augmenting of knowledge depth intensifies current knowledge base, technological domains, and R&D trajectories. In other words, increasing knowledge depth reinforces organisations' experience inertia (Yang, 2014; Yang *et al.*, 2014). Due to the sunk costs of current knowledge base, organisations tend to be risk-averse, and innovation trajectories of R&D teams may be locked with the augmenting of knowledge depth. Therefore, R&D staff tend to extend and reinforce current R&D trajectories with prior knowledge. As a result, organisations' exploitation performance gets better with the augmenting of knowledge depth.

As mentioned above, R&D capability has a positive impact on exploitation performance, and through the same mechanisms, knowledge boundary spanning of R&D network also promotes exploitation performance. However, will exploitation performance be better if corporations invest heavily in both kinds of resources in the meantime? The answer from us is no because the same mechanisms from both relationships may offset each other to some degree (Levinthal and March, 1993). We propose that for exploitation performance, investments in R&D capability can substitute investments in knowledge boundary spanning of R&D network. It is a waste of resources to invest heavily in the above two simultaneously. As a result, knowledge boundary spanning of R&D network negatively moderates R&D capability–exploitation performance relationship.

Figure 3 shows the theoretical framework of R&D capability–exploitation performance and knowledge boundary spanning of R&D network, and we propose that:

Hypothesis 2a: *Knowledge boundary spanning of R&D network negatively moderates the relationship between R&D capability and exploitation performance.*

The moderating role of knowledge boundary spanning of R&D network on R&D capability–exploration performance relationship from the horizontal dimension Knowledge boundary spanning of R&D network includes the spanning of knowledge breadth which means the augmenting of knowledge content

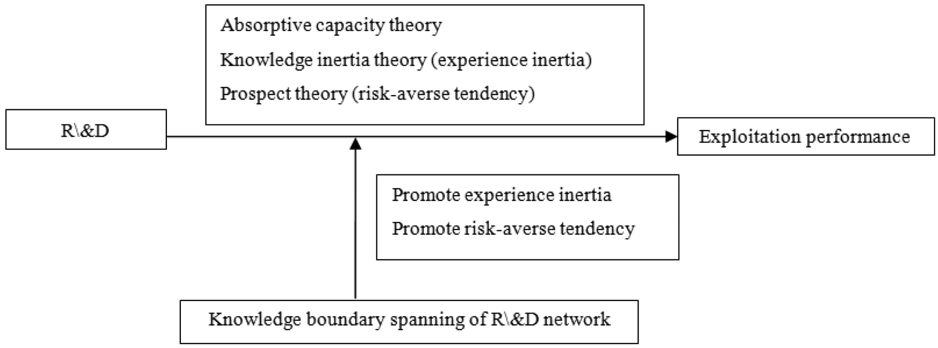

Fig. 3. Theoretical framework of R&D capability — exploitation performance and knowledge boundary spanning of R&D network.

heterogeneity. There is increase of knowledge diversity with knowledge boundary spanning of R&D network. Knowledge diversity contributes to generation of creative associations (Cohen and Levinthal, 1990) and novel ideas that are meaningful for exploration. If the level of knowledge heterogeneity is higher, there will be richer knowledge resources, knowledge sharing and knowledge conflicts (Xianan and Xudong, 2012). R&D teams are intellectual working teams who are open to conflicts, encourage doubts and differences, and advocate resolving conflicts through organisational learning (Huibin *et al.*, 2014). Knowledge boundary spanning of R&D network help organisations enhance organisational learning and increase knowledge heterogeneity. Therefore, learning inertia mentioned above can be overcome by knowledge breadth spanning. In accordance with the Medici Effect, inspirations or unexpected ideas can often be generated at junctions of different domains, thoughts and perceptions (Johansson, 2017). Teams with higher level of knowledge heterogeneity possess more abundant knowledge resources, broader comprehensive ability and more extensive external connections. Such R&D teams have capabilities to think and understand at a deeper level, adapt more quickly to changing environments and better identify opportunities for exploration (Huibin *et al.*, 2014). Therefore, knowledge boundary spanning for breadth overcomes learning inertia to some extent, promotes explorative learning and has a positive impact on explorative innovation outputs.

Levinthal and March (1993) argue that timely adjustments of aspirations and feedback of high value explorative innovation may break the curse of competency traps and form powerful organisational learning. The notion of intra-organisational knowledge boundary spanning mainly includes knowledge interaction between R&D staff and skilled staff. Skilled staff work on application of R&D outputs and communicate practice status with R&D staff (Xianxiang, 2009). Skilled staff is the pivot connecting R&D outputs and R&D commercialisation

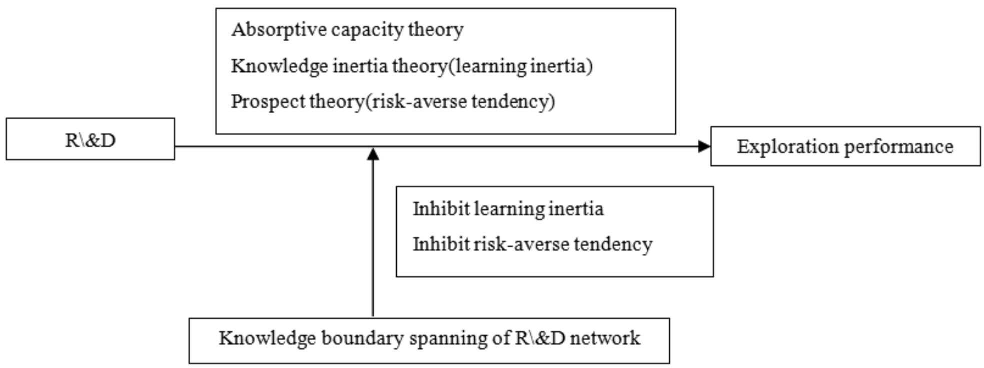

Fig. 4. Theoretical framework of R&D capability — exploration performance and knowledge boundary spanning of R&D network.

(Cohen and Levinthal, 1990). Due to the limitation of cognition, individuals always respond to environmental changes according to their own perception and cognition (Wang *et al.*, 2016). Feedback of different R&D outputs from skilled staff to R&D staff depict different kinds of commercialisation in detail. It adjusts previous aspirations of R&D teams for both kinds of innovation and promotes R&D teams to better understand exploitation's disadvantage of bringing short-term benefits and exploration's advantage of bringing high value and long-term returns (Levinthal and March, 1993). To some degree, knowledge boundary spanning for breadth overcomes risk-averse tendency caused by high level R&D capability and promotes R&D teams to carry on explorative innovation. In line with the above demonstrations, we propose that for exploration performance, investment in R&D capability complement investment in knowledge boundary spanning of R&D network. Investment in both kinds of resources can generate better exploration performance.

Figure 4 shows the theoretical framework of R&D capability–exploration performance and knowledge boundary spanning of R&D network, and we posit that:

Hypothesis 2b: *Knowledge boundary spanning of R&D network positively moderates the relationship between R&D capability and exploration performance.*

Methods

Sampling and data collection

To test our hypotheses, we examined corporations from the Hunan province applying for high-tech enterprises qualification in China. We collected application documents, excluded samples with incomplete data, and ultimately, obtained data of 702 corporations. The application documents consisted of R&D investment, firm assets, firm sales, the number of patents, utility models, scientific and technical staff, R&D staff, and others. All sample firms applying for high-tech enterprises qualification met the application standards and pursued stronger innovation capabilities, so the sample selection tallied well with the innovation subject of this study. In addition, high-tech enterprises are the key factors to stimulate economic growth and promote economic transformation in developing countries. Conducting studies on their innovation inputs and outputs are critical for the development of developing countries.

Measures

In order to reduce the influence of uncontrollable factors, this work examines hypotheses with sample data of 3 years (Wenjie and Ya, 2014). The data set used in this study covers the period between 2013 and 2015.

Exploitation performance: Utility models which are incremental R&D outputs tally well with attributes of exploitative innovation. In line with the study of (Beneito, 2006), the number of utility models registered during 2013–2014 was used as a measure of exploitation performance. Exploration performance: Patents which are radical innovation outputs tally well with attributes of explorative innovation. In line with the study of Beneito (2006), the number of patents registered during 2013–2014 was used as a proxy for exploration performance. According to World Intellectual Property Organisation (WIPO), utility models and patents are exclusive rights for holders to protect their commercial rights for a limited period of time. Utility models and patents cannot be used commercially without the right holders' authorisation. There are differences between utility models and patents. Firstly, the requirements to apply for patents are more stringent. Secondly, the duration of patents protection is longer than utility models. In China, the term of protection for patents is 20 years, and 10 years for utility models. Based on the above characteristics, Beneito (2006) take patents as significant innovation and utility models as incremental innovation.

R&D capability: We considered R&D investment for R&D department to approximate for R&D capability as Yong (2011) does in his study. Previous literatures show that R&D investments can promote organisational learning and strengthen capability to recognise proper technological opportunities and apply technological information from outside. In the meantime, R&D investments reinforce cultivation and accumulation of R&D capability (Stock *et al.*, 2001; Deeds *et al.*, 2000).

Knowledge boundary spanning of R&D network: We took the number of skilled staff as the proxy for knowledge boundary spanning of R&D network. Skilled human capital is a kind of innovative human capital but it is different from R&D human capital. Even though skilled staff do not participate in R&D activities directly, they are engaged in commercialisation of R&D outputs, and communicate practice processes with R&D staff to develop R&D knowledge network (Xianxiang, 2009). According to knowledge-based view, knowledge is embedded in individuals, thus to some degree, the number of individuals represents knowledge boundary spanning (Deeds *et al.*, 2000).

Control variables: We used logarithm of assets as an indicator of firm size. The effect of firm size on innovation performance mainly stems from the study of (Sehumpeter, 1942). Existing literatures have used sales, the number of employees and assets as proxy for firm size (Xu *et al.*, 2013). In this study, we chose assets as proxy of firm size. We referred to previous studies (Zhou and Wu, 2010; Mouse and Chowdhury, 2014) to control the firm's age and collect the age data manually. Ortega-Argilés *et al.* (2005) argue that a high degree of ownership concentration do not favour R&D outputs. Accordingly, we took ownership concentration as the control variable and measure it with the ownership percentage of the largest shareholder.

Empirical results

Table 1 shows the descriptive statistics of the constructs which consist of mean value, standard deviation, minimum value and maximum value.

Table 2 reports the correlation coefficients between all variables in this study. None of the correlation coefficients are large enough to indicate multicollinearity.

To test our hypotheses, we employed a stepwise hierarchical regression approach to assess the explanatory power of each set of variables (Aiken and West, 1994). When discussing models (M1–M6), the abbreviations for exploration performance (ERP), exploitation performance (EIP), R&D capability (RC) and

Table 1. Descriptive statistics of the constructs.

Constructs	Sample size	Mean	StdDev	Min	Max
Exploration performance (ERP)	702	1.32	2.38	0	23
Exploitation performance (EIP)	702	6.42	9.36	0	112
R&D capability (RC)	702	2755.21	14071.11	20.03	276596.30
Knowledge boundary spanning of R&D networks (KB)	702	68.03	337.45	0	7657
Firm age	702	10.61	6.33	0.58	62.92
Firm size	702	8.58	1.63	2.92	14.81
Ownership concentration	702	0.68	0.25	0.07	1

Table 2. Correlation matrix.

Constructs	ERP	EIP	RC	KB	Firm age	Firm size	Ownership concentration
ERP	1						
EIP	0.2651	1					
RC	0.3797	0.3698	1				
KB	0.2211	0.2902	0.6573	1			
Firm age	0.2121	0.2558	0.1699	0.1428	1		
Firm size	0.3080	0.3319	0.4028	0.3317	0.4088	1	
Ownership concentration	0.0053	0.0809	0.1063	0.097	0.1013	0.1908	1

knowledge boundary spanning of R&D network (KB) are used. The models (M1–M6) are as follows:

$$\begin{aligned}
\text{M1: EIP} &= \alpha_1 + \beta_1\ \text{Age}_i + \beta_2\ \text{Size}_i + \beta_3\ \text{Ownershipconcentration}_i + \varepsilon_i \\
\text{M2(H1a): EIP} &= \alpha_1 + \beta_1\ \text{Age}_i + \beta_2\ \text{Size}_i + \beta_3\ \text{Ownershipconcentration}_i \\
&\quad + \beta_4\ \text{RC}_i + \varepsilon_i \\
\text{M3(H2a): EIP} &= \alpha_1 + \beta_1\ \text{Age}_i + \beta_2\ \text{Size}_i + \beta_3\ \text{Ownershipconcentration}_i \\
&\quad + \beta_4\ \text{RC}_i + \beta_5\ \text{KB}_i + \beta_6\ \text{RC}_i\ \text{wKB}_i + \varepsilon_i \\
\text{M4: ERP} &= \alpha_1 + \beta_1\ \text{Age}_i + \beta_2\ \text{Size}_i + \beta_3\ \text{Ownershipconcentration}_i + \varepsilon_i \\
\text{M5(H1b): ERP} &= \alpha_1 + \beta_1\ \text{Age}_i + \beta_2\ \text{Size}_i + \beta_3\ \text{Ownershipconcentration}_i i \\
&\quad + \beta_4\ \text{RC}_i + \beta_5\ \text{RC}_i^2 + \varepsilon_i \\
\text{M6(H2a): ERP} &= \alpha_1 + \beta_1\ \text{Age}_i + \beta_2\ \text{Size}_i + \beta_3\ \text{Ownershipconcentration}_i i \\
&\quad + \beta_4\ \text{RC}_i + \beta_5\ \text{RC}_i^2 + \beta_6\ \text{KB}_i + \beta_7\ \text{RC}_i\ \text{wKB}_i \\
&\quad + \beta_8\ \text{RC}_i^2\ \text{wKB}_i + \varepsilon_i
\end{aligned}$$

We examined the hypotheses by ordinary least squares (OLS) regression. Table 3 reports the regression results of standard variables. With Hypothesis 1a, we considered the effect of R&D capability on exploitation. As M2 shows, the coefficient of RC ($b = 0.227$, $p < 0.01$) is positive and reaches the significance level, while R2 is 0.194. Therefore, we argue that R&D capability has a significantly positive impact on exploitation performance, in support of 1a. Hypothesis 1b deals with the relationship between R&D capability and exploration performance. As shown in M4, both RC and RC2 are not significantly associated with exploration performance. This indicates that the R&D capability does not have a significant U-shaped relationship with exploration performance so our data cannot support Hypothesis 1b.

In Hypothesis 2a, we assessed the moderating role of knowledge boundary spanning of R&D network on R&D capability–exploitation performance

Table 3. Standardised regression estimates[3].

	Exploitation performance M1	Exploitation performance M2	Exploitation performance M3	Exploitation performance M4	Exploitation performance M5	Exploitation performance M6
Firm age	0.142	0.141	0.116	0.0971	0.101	0.0955
	(0.130)	(0.130)	(0.128)	(0.0792)	(0.0756)	(0.0621)
Firm size	0.269***	0.159***	0.0916	0.256***	0.166***	0.0280
	(0.0621)	(0.0549)	(0.0588)	(0.0667)	(0.0649)	(0.0846)
Ownership concentration	0.0146	0.00614	−0.00174	−0.0537	−0.0612*	−0.0739**
	(0.0340)	(0.0324)	(0.0310)	(0.0366)	(0.0356)	(0.0338)
RC		0.277***	0.496**		0.143	1.025***
		(0.102)	(0.226)		(0.219)	(0.364)
RC_2					0.00946	−0.115***
					(0.116)	(0.0432)
KB			0.267***			0.369
			(0.103)			(0.332)
RC × KB			−0.0461*			−0.173**
			(0.0255)			(0.0785)
RC_2 × KB						0.0158***
						(0.00464)
Sample size	702	702	702	702	702	702
R_2	0.128	0.194	0.220	0.107	0.190	0.252

relationship. As M3 shows, the interaction of RC and KB ($b = -0.0461, p < 0.1$) has a significant negative effect on exploitation performance. Additionally, R2 increased from 0.194 in M2 to 0.220 in M3. Therefore, knowledge boundary spanning of R&D network negatively moderates the relationship between R&D capability and exploitation. This is in support of Hypothesis 2a. In Hypothesis 2b, we examined the moderating role of knowledge boundary spanning of R&D network on R&D capability–exploration relationship. As shown in M6, the first-order interaction between RC and KB ($b = -0.173$, $p < 0.05$) positively affects exploration performance whereas their second-order interaction ($b = 0.0158$, $p < 0.01$) negatively relates to exploration performance. Within the same model, the coefficient of RC2 ($b = -0.115$, $p < 0.01$) is significantly negative. The results also show that R2 increases from 0.190 in M5 to 0.252 in M6.

The effects of RC and RC2 on exploration performance are not significant in M4, but coefficients of RC, RC2, KB, the interaction of RC and KB, and the interaction of RC2 and KB are all significant with the addition of moderator and interactions. Why is this the case? What gives rise to the above situations? In order to further open the black box of exploration performance, R&D capability and knowledge boundary spanning of R&D network, we conducted a deeper analysis.

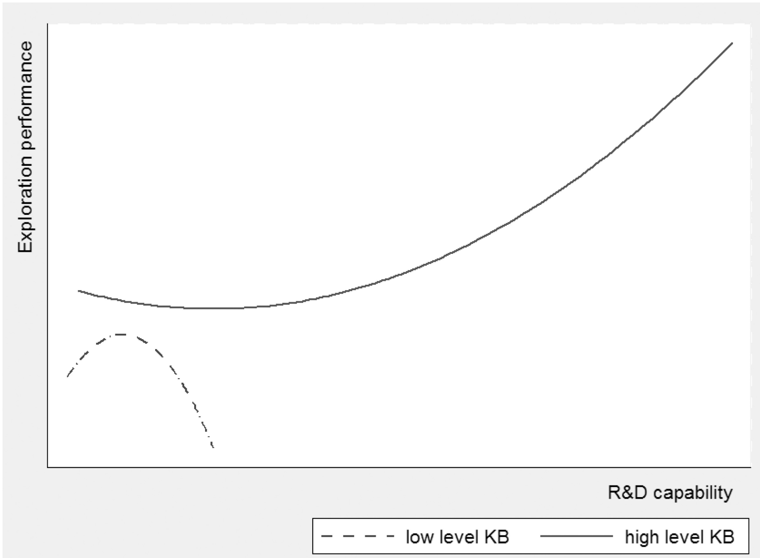

Fig. 5. The interaction between knowledge boundary spanning of R&D network and R&D capability.

According to Aiken and West (1994), the interaction of the moderator and the independent variable makes a difference in the curve line while the interaction of the moderator and quadratic term may change the opening direction of the curve line. As we can see from Fig. 5, when the knowledge boundary spanning of R&D network is a low 5, there is U-shaped relationship between R&D capacity and exploration performance. When the knowledge boundary spanning of R&D network reaches a high level 6, the slope is slower and the opening direction changes from downward to upward, so there is J-shaped relationship between R&D capability and exploration performance. Blundell *et al.* (1999) argue that skilled staff of higher level adapt to and accomplish new tasks in a shorter time, and are provided with important innovation sources. Our empirical results are consistent with the above viewpoints. Aiken and West (1994) argue that when testing a moderated quadrate relationship, the significance of the interaction between squared term and the moderator matters and determines if there is significant moderate effect. In this study, knowledge boundary spanning of R&D network changes the curvilinear relationship between R&D capability and exploration performance, and promotes exploration performance on an equal basis of R&D capability. Knowledge boundary spanning of R&D network positively moderates the relationship between R&D capability and exploration performance.

In addition, the results are robust when firm size is measured by sales instead of assets. As shown in Table 4, the empirical results are consistent with Table 3.

Table 4. Regression results for robustness test.

	Exploitation performance M1	Exploitation performance M2	Exploitation performance M3	Exploitation performance M4	Exploitation performance M5	Exploitation performance M6
Firm age	0.139	0.146	0.127	0.102	0.113	0.124*
	(0.142)	(0.142)	(0.139)	(0.0834)	(0.0808)	(0.0675)
Firm size	0.261***	0.141*	0.0608	0.251***	0.142***	−0.0508
	(0.0786)	(0.0734)	(0.0783)	(0.0698)	(0.0550)	(0.0696)
Ownership concentration	0.0242	0.0102	−0.000949	−0.0397	−0.0533	−0.0758**
	(0.0343)	(0.0328)	(0.0313)	(0.0384)	(0.0369)	(0.0359)
RC		0.286***	0.520**		0.178	1.265***
		(0.103)	(0.229)		(0.231)	(0.376)
RC_2					0.00944	−0.141***
					(0.0123)	(0.0454)
KB			0.282***			0.462
			(0.105)***			(0.348)
RC × KB			−0.0488*			−0.211***
			(0.0261)			(0.0812)
RC_2 × KB						0.0192***
						(0.00474)
Sample size	689	698	698	698	698	698
R_2	0.124	0.192	0.219	0.094	0.181	0.252

As M2 shows, the coefficient of RC ($b = 0.286$, $p < 0.01$) is positive and reaches the significance level, in support of Hypothesis 1a. As M4 shows, both RC and RC2 are not significantly associated with exploration performance. Our data cannot support Hypothesis 1b even when the variable firm size is measured by sales. As M3 shows, the interaction of RC and KB ($b = -0.0488$, $p < 0.1$) has a significant negative effect on exploitation performance in support of Hypothesis 2a. As shown in M6, the first-order interaction between RC and KB ($b = -0.211$, $p < 0.01$) positively affects exploration performance whereas their second-order interaction ($b = 0.0192$, $p < 0.01$) negatively relates to exploration performance. In addition, at the same model, the coefficient of RC2 ($b = -0.141$, $p < 0.01$) is significantly negative. Although the coefficients and significance levels are different from Table 3, the empirical results for the hypotheses are the same.

Discussions

Building on absorptive capacity, knowledge inertia and prospect theory, this study examines the R&D capability–innovation performance relationship and the moderating role of knowledge boundary spanning of R&D network. Our findings show that R&D capability has a significantly positive impact on exploitation performance but no significant impact on exploration performance. We also introduced the notion of knowledge boundary spanning of R&D network. As the empirical results show for exploitation performance, knowledge boundary spanning substitutes for R&D capacity. However, for exploration performance, they complemented each other. There were several contributions for existing literatures. This study chose the perspective of absorptive capacity, knowledge inertia and prospect theory to analyse R&D capability–innovation performance relationship, and examines if different levels of R&D capability promote or inhibit different kinds of innovation performance. This study argues that stronger R&D capability is in favour of exploitation performance because of stronger inertia and risk-averse tendency whereas stronger R&D capability inhibits exploration performance because of stronger learning inertia and risk-averse tendency. This is not reflected in the findings of this study. This study also extends to the literature of knowledge boundary spanning. Previous literatures of knowledge boundary spanning primarily pay attention to boundary spanning across corporations (Tortoriello and Krackhardt, 2010; Liu, 2015) but this study sheds light on boundary spanning within corporations (Rosenkopf and Nerkar, 2001). We also probed into boundary spanning from knowledge network of R&D staff to skilled staff within corporations. In detail, we deeply analyse cognition processes of innovation strategies and extend literatures of micro mechanism of innovation strategy management.

There are several contributions to management practices. Firstly, corporations should be aware that R&D capability has a positive impact on exploitation performance. More importantly, they should pay attention to limitations of R&D capability to exploration performance. For exploitation performance, knowledge boundary spanning of R&D network substitutes for R&D capability but for exploration performance, they complement each other. Additionally, different kinds of innovation have different influences on corporations in different development phases. Therefore, corporations should reconfigure resources according to their innovation strategy and allocate investments for R&D staff and skilled staff rationally. If corporations are inclined to pursue better exploitation innovation, they should either invest in R&D capability or knowledge boundary spanning as working along both lines can lead to a waste of resources. Corporations which are inclined to pursue excellent exploration innovation should pay attention to both R&D capability and knowledge boundary spanning. Corporations which emphasize only on the enhancement of R&D capability may fall into competency traps (Levinthal and March, 1993) and will always backfire. In short, investing in both kinds of resources can promote exploration performance. Pursuing equilibrium of exploitation and exploration, corporations should determine investment intensity of R&D capability and knowledge boundary spanning in accordance with practical situations. Secondly, high-tech enterprises are critical factors to promote economic growth in China (Yong, 2011). Investigations on their R&D capability and innovation performance can provide companies with references to develop and achieve innovation strategies scientifically, improve their innovation performance, and promote economic restructuring. Thirdly, China has enacted ⟨the 13th Five-Year Plan for Human Resources and Social Security Development⟩ for requirements of skilled talents building. ⟨The 13th Five-Year Plan⟩ involves cultivation, evaluation and motivation of skilled talents hoping to foster sufficient, well-constructed and highly qualified skilled staff in order to promote the transformation and upgrading of Made in China. We regard skilled staff from the view of knowledge boundary spanning of R&D network and examine its substitutional relation or complementary relation with R&D capability. Empirical results show that skilled staff of high level boost exploration performance. In light of this, the present study provides with references for skilled staff development.

There are several limitations of this study. Firstly, it is limited by available data. Our empirical research is restricted to cross-sectional data, with uncontrollable influences on time factor. Secondly, although we support hypotheses through absorptive capacity, knowledge inertia and prospect theory, we do not directly quantify these elements. Further studies may excavate the above elements at a deeper level, collect more elaborate data to support and open the black box of R&D capability, exploitation performance and exploration performance further.

References

Afuah, A (2002). Mapping technological capabilities into product markets and competitive advantage: The case of cholesterol drugs. *Strategic Management Journal*, 23(2), 171–179.

Aiken, LS and SG West (1994). Multiple regression: Testing and interpreting interactions. *Journal-Operational Research Society*, 45, 119–119.

Ayhan, MB and E Oztemel (2011). A methodology for measuring product innovation: A case study for a manufacturing system. *International Journal of Manufacturing Technology and Management*, 24(1–4), 139–152.

Beneito, P (2006). The innovative performance of in-house and contracted R&D in terms of patents and utility models. *Research Policy*, 35(4), 502–517.

Blundell, R, L Dearden, C Meghir and B Sianesi (1999). Human capital investment: The returns from education and training to the individual, the firm and the economy. *Fiscal studies*, 20(1), 1–23.

Boer, H and WE During (2001). Innovation, what innovation? A comparison between product, process and organisational innovation. *International Journal of Technology Management*, 22(1–3), 83–107.

Cohen, WM and DA Levinthal (1990). Absorptive capacity: A new perspective on learning and innovation. *Administrative Science Quarterly*, 128–152.

Deeds, DL, D DeCarolis and J Coombs (2000). Dynamic capabilities and new product development in high technology ventures: An empirical analysis of new biotechnology firms. *Journal of Business Venturing*, 15(3), 211–229.

Dougherty, D and T Heller (1994). The illegitimacy of successful product innovation in established firms. *Organization Science*, 5(2), 200–218.

Gunday, G, G Ulusoy, K Kilic and L Alpkan (2011). Effects of innovation types on firm performance. *International Journal of Production Economics*, 133(2), 662–676.

Huibin, Y and Y Daming, Central South University (2014). An empirical study on the impact of knowledge conflict of R&D team on radical innovation performance of enterprises. Chinese Journal of Management.

Johansson, F (2017). *The Medici Effect, With a New Preface and Discussion Guide: What Elephants and Epidemics Can Teach Us About Innovation?* Massachusetts: Harvard Business Review Press.

Kahneman, D (1979). Prospect theory: An analysis of decisions under risk. *Econometrica*, 47, 278.

Levinthal, DA and JG March (1993). The myopia of learning. *Strategic Management Journal*, 14(S2), 95–112.

Liao, SH (2002). Problem solving and knowledge inertia. *Expert Systems with Applications*, 22(1), 21–31.

Liao, SH, WC Fei and CT Liu (2008). Relationships between knowledge inertia, organizational learning and organization innovation. *Technovation*, 28(4), 183–195.

Liu, Y (2015). R&D network boundary spanning, knowledge base, and innovation catching-up. *Study in Science of Science*, 33(1), 916–923.

March, JG (1991). Exploration and exploitation in organizational learning. *Organization Science*, 2(1), 71–87.

Meng, WXY (2015). Semantic modeling for knowledge exchange networks of R&D staff in an open innovation environment. *Information Science* (10), 36–40.

Ooi, KB, B Lin, PL Teh and AYL Chong (2012). Does TQM support innovation performance in Malaysia's manufacturing industry? *Journal of Business Economics and Management*, 13(2), 366–393.

Organisation for Economic Co-operation and Development (1997). *The Measurement of Scientific and Technological Activities: Proposed Guidelines for Collecting and Interpreting Technological Innovation Data: Oslo Manual*. OECD.

Ortega-Argilés, R, R Moreno and JS Caralt (2005). Ownership structure and innovation: Is there a real link? *The Annals of Regional Science*, 39(4), 637–662.

Paruchuri, S and M Eisenman (2012). Micro foundations of firm R&D capabilities: A study of inventor networks in a merger. *Journal of Management Studies*, 49(8), 1509–1535.

Rosenkopf, L and A Nerkar (2001). Beyond local search: Boundary-spanning, exploration, and impact in the optical disk industry. *Strategic Management Journal*, 22(4), 287–306.

Rui, L (2016). An empirical study on the impact of R&D investment and employee involvement on innovation commercialization: Evidence from it firms in china. *Journal of Modern Accounting and Auditing*, 12(8), 429–442.

Sehumpeter, JA (1942). *Capitalism, Socialism and Democracy*. New York: Hamper Brother. 連結.

Seibert, SE, ML Kraimer and RC Liden (2001). A social capital theory of career success. *Academy of Management Journal*, 44(2), 219–237.

Still, MC and D Strang (2009). Who does an elite organization emulate? *Administrative Science Quarterly*, 54(1), 58–89.

Stock, GN, NP Greis and WA Fischer (2001). Absorptive capacity and new product development. *The Journal of High Technology Management Research*, 12(1), 77–91.

Swift, T (2016). The perilous leap between exploration and exploitation. *Strategic Management Journal*, 37(8), 1688–1698.

Tijssen, RJ (1998). Quantitative assessment of large heterogeneous R&D networks: The case of process engineering in the Netherlands. *Research policy*, 26(7), 791–809.

Tortoriello, M and D Krackhardt (2010). Activating cross-boundary knowledge: The role of Simmelian ties in the generation of innovations. *Academy of Management Journal*, 53(1), 167–181.

Wang, C, C Wang, B Luo, B Luo, Y Liu, Y Liu and Z Wei (2016). The impact of executives' perceptions of environmental threats and organizational slack on innovation strategies. *Nankai Business Review International*, 7(2), 216–230.

Wenjie, JJW and S Ya (2014). Research on entrepreneurs' social network, dualistic innovation and technological innovation performance. *Science Research Management*, 35(2), 43–53.

Xianan, L and N Xudong (2012). A knowledge integrated research based on team knowledge heterogeneity structure. *Science & Technology Progress and Policy*, 11, 132–137.

Xianxiang, K (2009). Study on classification of innovative human capital. *Science and Technology Management Research*, 29(7), 335–337.

Xu, S, F Wu and E Cavusgil (2013). Complements or substitutes? Internal technological strength, competitor alliance participation, and innovation development. *Journal of Product Innovation Management*, 30(4), 750–762.

Yang, L (2014). Organizational heuristics, R&D network boundary spanning, and innovation catch-up of latecomer firms in a transition economy, PhD thesis, Zhejiang University, China.

Yang, KP, C Chou and YJ Chiu (2014). How unlearning affects radical innovation: The dynamics of social capital and slack resources. *Technological Forecasting and Social Change*, 87, 152–163.

Yang, L, W Jiang and J Shisong (2013). How do the latecomer firms catch up in innovation? A study from the perspective of the boundary expanding in R&D. *Management World*, 3.

Yong, D (2011). R&D capability, scale and high technology enterprises' performance. *Nankai Economic Studies*, 4, 011.

Zheng, X (2013). The impact of enterprise human capital on technological innovation. PhD thesis, Hebei University of Technology, China.
Zhou, KZ and CB Li (2012). How knowledge affects radical innovation: Knowledge base, market knowledge acquisition, and internal knowledge sharing. *Strategic Management Journal*, 33(9), 1090–1102.
Zhou, KZ and F Wu (2010). Technological capability, strategic flexibility, and product innovation. *Strategic Management Journal*, 31(5), 547–561.